

LAB ASSIGNMENT - 04

Aim: Write a Java program to showcase the use of File I/O Exception Handling in Java.

Objectives: i) To study I/O stream classes.
Understand the different types of exceptions.
Exception handling using various Exception classes.

Theory:

Java I/O: input-output stream classes:

It is used to read & write data.

Types of streams:-

Byte stream class: Used for handling binary data.

Character stream classes: Used for handling text data.

Explain exception handling mechanism in Java.

Exception handling is a mechanism to handle runtime errors so that the program continues execution.

Keywords used: try → code that may cause exception.

catch → handles exception

finally → always executes

throw → manually throw exception.

throws → declares exception.

Write types of exception classes:

Checked exceptions: (Compile time).

→ Must be handled. Eg: IO exception, SQL exception etc

b) Unchecked exceptions (Runtime):

→ Occur at runtime. Eg → null pointer exception, arithmetic exception

• Algorithm:

- i) Start
- ii) Accept customer details (cid, cname, amount)
- iii) Validate : a) cid between 1 & 20
b) amount ≥ 1000
c) amount should be positive.
- iv) If invalid → throw user-defined exception.
- v) Display menu: Deposit ; Withdraw.
- vi) For withdrawl: If withdrawl amount > balance → throw exception.
- vii) Store valid records into file.
- viii) Handle exceptions using try-catch
- ix) End.

• Platform: Eclipse, NetBeans, VSCode, IntelliJ.

• Conclusion: Thus, studied the exception handling concepts in java & usage of I/O stream classes.

• FAQs:

Q.1 What is the difference between error & exception?

Error

Serious problem in system.
Can't be handled by programmer.
Occurs at JVM level.
Not recoverable.

Exception

- i) Problem in program logic.
- ii) Can be handled using try-catch.
- iii) Occurs in application.
- iv) Recoverable.

a few

Write examples for checked & unchecked exceptions:

Checked exceptions must be handled using try-catch or throws.

Examples: IOException, SQLException.

Eg: try {

```
    FileReader fr = new FileReader("file.txt");  
    } catch (FileNotFoundException e) {  
        System.out.println("File not found");  
    }
```

→ Unchecked exceptions → occur at runtime & aren't checked at compile time.

Eg: ArithmeticException, NullPointerException.

Eg code: `int a = 10/0; // ArithmeticException.`

3 What are different keywords in exception handling in Java?

- i) try: Contains code that may cause an exception.
- ii) catch: Handles the exception.
- iii) finally: Always executes (used for cleanup like closing files)
- iv) throw: Used to explicitly throw an exception.
- v) throws: Declares exceptions in method signature.

Q.4 What is the difference between throws & throw keywords in Java?

throw	throws
i) Used to throw exception	i) Declares exception.
ii) Inside method body.	ii) In method signature.
iii) No. of exceptions → one at a time.	iii) No. of exceptions → can declare multiple.
iv) Keyword position → Within code block.	iv) Keyword position → After method name.

Q.5 List down various Byte & Character Stream Classes in Java.

Ans: a) Byte stream classes :-

→ Used for binary data like images, audio, etc.

Common classes: File InputStream, File OutputStream etc.

b) Character Stream classes :-

→ Used for text data (characters)

Common classes: File Reader, File Writer.